

Xiyuan Yang

Computer Science Ph.D. Student @ University of Illinois Urbana-Champaign

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Research & Selected Experiences

- My research focuses on agentic foundation models, including: **AutoResearch**, **World Models** ([arXiv 2026](#)), **Optimization** ([ICML 2026](#)) and **Trustworthy Agents** ([EMNLP 2025](#), [CVPR 2024](#)).
- Have implemented large scale agentic-RL pipelines with **fully async** rollout / training loops, scaling multi-turn, long-horizon (**64k+ tokens traj.**) training across **8+ nodes** on **~10B dense and ~30B MoE** models.

Education

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|---|----------------|
|  Ph.D. in Computer Science | 2025-2030 |
| University of Illinois Urbana-Champaign | Champaign, USA |
| Advisor: Prof. Jingrui He | |
|  B.Eng. in Computer Science and Technology | 2021-2025 |
| Wuhan University | Wuhan, China |
| Advisor: Prof. Mang Ye | |

Internship Experience

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|---|----------------|
|  Google, Google Cloud | 2026-2027 |
| Mentor: Shentao Yang | Sunnyvale, USA |
| Project: Extra Long-Horizon RL (e.g. 1M+ token ctx) for Agentic Foundation Models | |
|  Amazon, Prime Video Group | 2026 |
| Mentor: Zhenyu Liao | Sunnyvale, USA |
| Project: Scaling Automatic Research Agents via World Models | |
|  Microsoft Research, Social Computing Group | 2024-2025 |
| Mentor: Fangzhao Wu | Beijing, China |
| Project 1: Detecting Prompt Injections via Representation Engineering (in EMNLP 2025) | |
| Project 2: Identifying Cross-lingual Inequalities of Foundation Models (in PNAS 2025) | |
| Project 3: Defending Jailbreak Attacks by Vision Language Models (under review of Nature Communications) | |
| Project 4: Measuring Human Contributions in AI-Generated Content (in ACL 2026) | |
| Competition : Foundation Model based News Recommendation (Ranked 1/157 groups) | |
|  University of Chicago, Department of Computer Science | 2024 |
| Mentor: Prof. Tian Li | Chicago, USA |
| Project: Differentially Private Distributed Optimization (under review) | |

Invited Presentations

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|  Jump Trading AI Research Symposium (selected poster, ~25 worldwide) | 2026 |
| Optimizer Aware Kernels for Foundation Model Training Dynamics | New York City, USA |

Publications

- **Scaling Automatic Research Agents via World Models**
Xiyuan Yang, Sheikh Sarwar, Jingru Cheng, Zhan Shi, Duanshun Li, Huiyuan Chen, Haiyang Zhang, Chenlei Guo, Jingrui He, Zhenyu Liao
arXiv 2026
- **Code as Agent Harness** [Huggingface Daily Paper #1]
Xuying Ning et al. (43 authors, incl. Xiyuan Yang)
arXiv 2026
- **Heterogeneous Scientific Foundation Model Collaboration** [Huggingface Daily Paper #1]
Zihao Li, Jiaru Zou, Feihao Fang, Xuying Ning, Mengting Ai, Tianxin Wei, Sirui Chen, Xiyuan Yang, Jingrui He
arXiv 2026
- **Preconditioning Neural Tangent Kernel for Adaptive Optimization**
Xiyuan Yang, Wenxuan Bao, Katherine Tieu, Jingrui He
ICML 2026
- **Latent Collaboration in Multi-Agent Systems** [Huggingface Daily Paper #1]
Jiaru Zou, Ruizhong Qiu, Gaotang Li, Xiyuan Yang, Katherine Tieu, Pan Lu, Ke Shen, Hanghang Tong, Yejin Choi, Jingrui He, James Zou, Mengdi Wang, Ling Yang
ICML 2026
- **Ramen: Robust Test-Time Adaptation of Vision-Language Models with Active Sample Selection**
Wenxuan Bao, Yanjun Zhao, Xiyuan Yang, Jingrui He
CVPR 2026
- **Measuring Human Contribution in AI-Assisted Content Generation**
Yueqi Xie, Tao Qi, Jingwei Yi, Xiyuan Yang, Ryan Whalen, Junming Huang, Qian Ding, Yu Xie, Xing Xie, Fangzhao Wu
ACL 2026
- **Defending LLMs Against Jailbreak Attacks Utilizing Cross-Modality Generalization Gap**
Xiyuan Yang*, Chenglong Wang*, Haoyu Tang*, Yueqi Xie, Bin Zhu, Lingjuan Lyu, Mang Ye, Fangzhao Wu
Under Review 2025
- **Uncovering Inequalities in New Knowledge Learning by Large Language Models Across Different Languages**
Chenglong Wang, Haoyu Tang, Xiyuan Yang, Yueqi Xie, Jina Suh, Sunayana Sitaram, Junming Huang, Yu Xie, Pengjun Zhao, Zhaoya Gong, Xing Xie, Fangzhao Wu
PNAS 2025
- **Defending Against Indirect Prompt Injection by Instruction Detection**
Tongyu Wen*, Chenglong Wang*, Xiyuan Yang*, Haoyu Tang, Yueqi Xie, Lingjuan Lyu, Zhicheng Dou, Fangzhao Wu
EMNLP 2025
- **Differentially Private Federated Clustering with Random Rebalancing**
Xiyuan Yang, Shengyuan Hu, Soyeon Kim, Tian Li
Under Review 2025

- **FedAS: Bridging Inconsistency in Personalized Federated Learning**
Xiyuan Yang, Wenke Huang, Mang Ye
CVPR 2024
- **Dynamic Personalized Federated Learning with Adaptive Differential Privacy**
Xiyuan Yang*, Wenke Huang*, Mang Ye
NeurIPS 2023
- **Robust Heterogeneous Federated Learning under Data Corruption**
Xiuwen Fang, Mang Ye, Xiyuan Yang
ICCV 2023

Community Contributions

- **PFLlib**: Personalized and Distributed Optimization Library Github Stars **2000+**
- **LatentMAS**: Latent Communication of Multi-agent System Github Stars **600+**
- **Conference Review**: NeurIPS, ICML, ICLR, AAAI, AISTATS, CVPR, ICCV
- **Journal Review**: IEEE TPAMI, IEEE TMC, IEEE TNNLS, IEEE TIFS, IEEE TKDE, IEEE TDSC

Awards & Scholarships

- PhD Student Fellowship (first-year), University of Illinois Urbana-Champaign 2025
- Star of Tomorrow Award, Microsoft Research 2025
- Overseas Exchange and Study Award, Wuhan University 2024
- Leijun Computer Science Research Award, Wuhan University 2024

Skills

- **Machine Learning**: PyTorch, CUDA, HuggingFace Transformers, vLLM, NumPy
- **Systems & Tools**: Linux, Distributed Systems, Docker, Git, LaTeX